MILITARY SPECIFICATION

DIISOCYANATE, DIMERYL, C_{36} ALIPHATIC, TECHNICAL (DDI)

This specification has been approved by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the procurement of one type of C_{36} aliphatic, dimeryl, diisocyanate hereinafter referred to as DDI.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

WS 4503B	Rocket Motor, Mk 56 Mods 0, 1 & 2

STANDARDS

FEDERAL

FED-STD-313	Material Safety Data Sheets; Preparation and Submission of

MILITARY

MIL-STD-129	Marking for Shipment and Storage

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (Code Ident 10001)

OD 18856	Diisocyanate Compounds, Determination of
OD 18869	Density of Liquids, Determination of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1544	Standard Method of Test for Color of Transparent Liquids (Gardner Color Scale)
ASTM D 1638	Standard Methods of Testing Urethane Foam Isocyanate Raw Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

CODE OF FEDERAL REGULATIONS

49 CFR 100-199	Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)").

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS.

3.1 General description. The DDI is a clear, amber, low viscosity liquid. DDI has the following chemical structure:

$$O = C = N - [D] - N = C = O$$

where D is a 36 carbon, hydrocarbon radical.

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3.2 Form. The DDI shall be a liquid at 25°C.

3.3 Uniformity. The uniformity (homogeneity) of the DDI shall be such that the difference in isocyanate content between any two samples from a lot is not greater than 0.05 milliequivalents per gram (meq/g).

3.4 Chemical and physical properties. The chemical and physical properties of the DDI shall be in accordance with TABLE I.

TABLE I. Chemical and physical properties.

Characteristic	Minimum	Maximum
Isocyanate content, meq/g	3.25	3.41
Hydrolyzable chloride, wt%		0.05
Color, Gardner		7
Density, 25°C, g/cm^3	0.917	0.931

3.5 Product and process changes. The contractor shall notify the procuring activity of any changes in the product or processing techniques.

3.5.1 Mark 56 Mod 2 Dual Thrust Rocket Motor procurement. When the diisocyanate is procured for the Mk 56 Mod 2 Dual Thrust Rocket Motor, the requirements for change of material, material vendors, processes or equipment shall be those in WS 4503, paragraph 3.1.3 (6.2.1).

3.6 Material safety data sheets. The contractor shall prepare and submit material safety data sheets in accordance with FED-STD-313 as specified in the contract.

3.7 Workmanship. The material shall be uniform, free from contamination, foreign material or any other defect that would prevent its use for the purpose intended. The material shall be manufactured in accordance with standard manufacturing procedures of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspection shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

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4.3 Certificate of compliance. When specified in the contract (see 6.2), the contractor shall furnish a certificate of compliance stating that the DDI is in conformance with this specification.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the tests and examinations specified in TABLE II. Tests for iscyanate content (3.4) and uniformity (3.3) shall be performed on each sample selected in accordance with 4.4.1. All other tests and examinations shall be performed in duplicate on one or more samples selected in accordance with 4.4.1.

4.4.1 Sampling. Four containers shall be randomly selected from each lot and shall be individually sampled. If the lot consists of less than four containers, each of the containers shall be sampled. Each sample shall be placed in a separate sampling jar.

4.4.2 Lot. A lot shall consist of all DDI manufactured by the same process, from the same raw materials, by one manufacturer, at one plant under essentially identical conditions and to be offered for acceptance at one time. Several batches or parts thereof, manufactured from the same raw materials may be blended to form a uniform larger batch which shall then constitute a lot for inspection purposes.

4.4.3 Rejection criteria. If any individual sample does not conform to the requirements of this specification, the lot shall be rejected.

TABLE II. Quality conformance inspection.

Test	Requirement	Test Method
Packaging and marking	Section 5	Visual
Form	3.2	Visual
Uniformity	3.3	Comparison of values from 4.5.1
Isocyanate content	3.4	4.5.1
Hydrolyzable chloride	3.4	ASTM D 1638, Sect.40 to 46
Color	3.4	ASTM D 1544
Density	3.4	4.5.2
Workmanship	3.7	Visual

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4.5 Test methods.

4.5.1 Isocyanate content. The isocyanate content of the material shall be determined in accordance with OD 18856, except the sample size shall be 2.5 to 3.5 grams and the isocyanate content shall be calculated as follows:

$$\text{Isocyanate content, meq/g} = \frac{[V_1 - V_2]\,N}{W}$$

4.5.2 Density. The density of the DDI shall be determined in accordance with OD 18869, except the sample of material to be tested is the DDI specified herein at a test temperature of 25°C.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract (see 6.2), packaging and packing shall be level C.

5.1.1 Level C. Unless otherwise specified in the contract (see 6.2), packaging and packing of DDI shall be in accordance with standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination for one year from the date of manufacture when stored under the conditions specified in 6.5. Container size shall be as specified in the contract (see 6.2). Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-179. Marking shall include but not be limited to, the following information:

a. Title, number and date of this specification

b. Manufacturer's name and location

c. Material trade name.

d. Net weight and volume.

e. Lot number, batch number(s), and date of manufacture.

f. Storage conditions.

g. Toxicity precautions, if applicable.

h. Contract number.

i. Warranty provisions (see 6.2).

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and contents thereof shall be reinspected prior to shipment of ... closures shall be repaired or replaced

6. NOTES

6.1 Intended use. C_{36} Aliphatic Diisocyanate (DDI) is intended to be used as a propellant ingredient for the mark 56 Mod 2 Dual Thrust Rocket Motor or other rocket motor applications.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Quantity required.

c. Place of delivery.

d. Inspection conditions when other than as specified (see 4.2).

e. Packaging requirements if other than as specified (see 5.1).

f. Size of container required (see 5.1.1).

g. Any special marking required (see 5.2).

h. Warranty requirement (see 6.7)

i. Changes to material, material vendors, processes and equipment (see 3.5.1).

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID*
4.3	Certificate of compliance	DI-E-2121

*DIDs (Data Item Description/DD Form 1664) for the above requirements are documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Rounding-off specified limits. The following applies to all specified limits in this specification. For purposes of determining conformance with these requiremnts, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29, Standard Recommended Practice for Indicating Which Places of Figures are to be Considered Significant in Specified Limiting Values.

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6.4 Toxicity and safety. The material is substantially less toxic than diisocyanates commonly available. Diisocyanates are generally direct irritants and can be allergens. This material is nonsensitizing. Avoid contact with the skin, eyes and clothing.

6.5 Storage conditions. The DDI should be stored in well sealed containers at a maximum temperature of 27°C (80°F).

6.6 Shelf life. The shelf life of the material is normally at least 12 months from date of manufacture when stored in the original sealed containers under the conditions of 6.5. The DDI should be inspected prior to use. A skin on the material surface is indication of unacceptability.

6.7 Warranty. The contract should require a 12 month shelf life (see 6.2) warranty in accordance with the Armed Services Procurement Regulation (ASPR 1-324).

6.8 Suggested source of supply. A product that has met the requirements of this specification in past procurement actions is DDI no. 1410 manufactured by General Mills Chemicals Inc., P.O. Box 191, South Kensington Rd., Kankakee, IL 60901 (FMCM 11884). This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

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the solicitation (see 6.7)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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